



INSTITUTE OF INNOVATION

School Of Technology

Programming Principles and Practice with C and C++

Total Marks: 50

Instructions

1. All 28 questions are compulsory – 25 MCQ and 3 Programming
2. The Multiple Choice Question (MCQ) section carries a total of 25 marks, with each question allotted 1 mark.
3. The Programming section carries a total of 25 marks, distributed as 5 marks, 8 marks, and 12 marks respectively.
4. No negative marking will be applied in any section.

SECTION-A (Q1 – Q10)

(1 mark each)

Question 1

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int s = 0, n = 8;
    for (int i = 1; i <= n; ++i)
        for (int j = 1; j <= n; ++j)
            s += i + j;
    printf("%d\n", s);
    return 0;
}
```

- A. 512
 - B. 576
 - C. 640
 - D. 592
-

Question 2

What is the output of the following code snippet?

```
#include <stdio.h>
int main() {
    char c = 'd';
    int x = 70;
    printf("%d", c + x);
}
```

- A. 170
 - B. 168
 - C. 169
 - D. 171
-

Question 3

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    signed char c = 259;
    printf("%d\n", (int)c);
    return 0;
}
```

- A. 3
 - B. 1
 - C. 259
 - D. Implementation-defined / varies
-

Question 4

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    char a = 20, b = 30;
    char d = a * b;
    short x = a * b - d / 100 - 500;
    printf("%d\n", x);
    return 0;
}
```

- A. -450
 - B. -470
 - C. 100
 - D. 600
-

Question 5

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    signed char a = 80;
    signed char b = 100;
    signed int c = a + b;
```

```
    printf("%d\n", (int)c);  
    return 0;  
}
```

- A. 180
 - B. -76
 - C. 9216
 - D. 44
-

Question 6

What is the output of the following code snippet?

```
#include <stdio.h>  
int main(void) {  
    unsigned short a = 50;  
    signed short b = -10;  
    printf("%d\n", a > b + 30);  
    return 0;  
}
```

- A. 1
 - B. 0
 - C. -1
 - D. Compile error
-

Question 7

What is the output of the following code snippet?

```
#include <stdio.h>  
int main(void) {
```

```
int a = 5, b = 7, c = 2, d = 6, z = 0;
z = a+++ ++b*c--- --d;
printf("%d\n", z);
return 0;
}
```

- A. 41
 - B. 39
 - C. 37
 - D. 16
-

Question 8

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int a, b, c, d;
    a = b = c = d = 3;
    a *= b + 2;
    c += d *= 2;
    printf("%d\n", a + c);
    return 0;
}
```

- A. 24
 - B. 21
 - C. 19
 - D. 25
-

Question 9

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int a = 7;
    a = a++;
    a = a++ * a--;
    int c = a;
    int d = a++ * a++;
    int x = (c + d) / 100;
    printf("%d\n", x);
    return 0;
}
```

- A. 0
 - B. 10
 - C. 23
 - D. 32
-

Question 10

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int a, b, c, p = 15;
    p += (a = 6, b = a + 3, c = a + b);
    printf("%d\n", p);
    return 0;
}
```

A. 30

B. 31

C. 29

D. 32

SECTION-B (Q11 – Q20)

(1 mark each)

Question 11

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int i = -2;
    unsigned int j = 1;
    int x = 0;
    if (i < j) x = 10;
    else x = 20;
    printf("%d\n", x);
    return 0;
}
```

A. 10

B. 20

C. Compile error

D. Undefined behavior

Question 12

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
```

```
int a = 3, y = 10;
switch (a) {
    y = 30;
case 2:
    y++;
    break;
case 3:
    y--;
    break;
default:
    y += 5;
}
printf("%d\n", y + 1);
return 0;
}
```

- A. 10
 - B. 31
 - C. 29
 - D. 32
-

Question 13

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int v = 3, a = 1, b = 3;
    switch (v) {
        case a:
            a++;
            break;
```



```
        case b:
            b++;
            break;
    }
    printf("%d\n", a + b);
    return 0;
}
```

- A. 4
 - B. 5
 - C. 3
 - D. Error
-

Question 14

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int g = 9, h = 0;
    if ((g / 2) == 4) h = h + 1;
    if ((g % 4) == 1) h = h + 2;
    if ((g / 2) != 4) h = h + 5;
    printf("%d\n", h);
    return 0;
}
```

- A. 1
- B. 3
- C. 8
- D. 7

Question 15

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int i, j, k = 0, c = 0;
    j = 2 * 4 / 5 + 3.0 / 6 + 9 / 4;
    k -= --j;
    for (i = 0; i < 4; i++) {
        switch (i + k) {
            case 0: case 1: c += 1;
            case 2: ++c;
            default: c++;
        }
    }
    printf("%d\n", c);
    return 0;
}
```

- A. 10
 - B. 13
 - C. 11
 - D. 8
-

Question 16

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int i, j, count = 0;
```

```

i = 0;
for (j = -2; j <= 2; j++) {
    if ((j >= 0) && (i++)) count += j;
}
count += i;
printf("%d\n", count + i);
return 0;
}

```

- A. 7
- B. 5
- C. 9
- D. 4

Question 17

What is the output of the following code snippet?

```

#include <stdio.h>
int main(void) {
    int a = 3, b = -2, c = 4, d = -1, x = 0;
    if (a = b - c - d) x = (a++, b--, c++);
    else x = (c--, ++a, ++b);
    printf("%d\n", x);
    return 0;
}

```

- A. 0
- B. 4
- C. 2
- D. Compile-time error

Question 18

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    unsigned char i = 254;
    while (i++ != 0);
    printf("%d\n", i);
    return 0;
}
```

- A. 0
 - B. 1
 - C. 255
 - D. 2
-

Question 19

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int i = 1, j = 0, s = 0, n = 8;
    for (i = 1; i <= n; i++)
        for (j = i; j <= i; j++) s = s + j;
    printf("%d\n", s);
    return 0;
}
```

- A. 36
- B. 32

- C. 28
 - D. 45
-

Question 20

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int i = 0, k = 0, j = 3;
    i = j < k ? k : j--;
    if (j < i) j = j + k - 1;
    if (j == i) j = j - i;
    else j = j + --k;
    printf("%d\n", j + k - i + 7);
    return 0;
}
```

- A. 4
 - B. 5
 - C. 3
 - D. 6
-

SECTION-C (Q21 – Q25)

(1 mark each)

Question 21

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    unsigned long long c = 0, n = 1ULL << 20; // 2^20
    while (n) {
```

```

        c++; n >>= 1;
    }
    printf("%llu\n", c);
    return 0;
}

```

- A. 20
- B. 21
- C. 22
- D. 19

Question 22

What is the output of the following code snippet?

```

#include <stdio.h>
int main(void) {
    int i, j, k = 0;
    j = 3 * 4 / 5 + 2.0 / 4 + 7 / 3;
    k -= --j;
    int c = 0;
    for (i = 0; i < 4; i++) {
        switch (i + k) {
            case 0:
            case 1: ++c;
            case 2: ++c;
            default: ++c;
        }
    }
    printf("%d\n", c);
    return 0;
}

```

- A. 6
 - B. 8
 - C. 10
 - D. 12
-

Question 23

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int a = 15, b = 27, x = 0;
    a = b++ + a++;
    b = ++b + ++a;
    int sum = a + b - 19;
    for (int i = sum; i >= 1; i /= 2) x += 1;
    printf("%d\n", x);
    return 0;
}
```

- A. 3
 - B. 4
 - C. 6
 - D. 7
-

Question 24

What is the output of the following code snippet?

```
#include <stdio.h>
int main(void) {
    int a = 0, i, j, k, n = 7;
```

```
    for (i = 1; i <= n; i++)  
        for (j = 1; j <= i; j++)  
            for (k = 1; k <= j; k++)  
                a = a + 1;  
    printf("%d\n", a);  
    return 0;  
}
```

- A. 140
 - B. 120
 - C. 84
 - D. 56
-

Question 25

What is the output of the following code snippet?

```
#include <stdio.h>  
int main(void) {  
    int a = 0, i, j, k, n = 8;  
    for (i = 1; i <= n; i++)  
        for (j = 1; j <= i; j++)  
            for (k = 1; k <= j; k++)  
                a = a + 1;  
    printf("%d\n", a);  
    return 0;  
}
```

- A. 120
- B. 165
- C. 220
- D. 84

Section-D (Q26-Q28) Programming Questions

Question 26

(5 marks)

Write a C program to determine if a year is a leap year using an if-else-if ladder.

Input Example

2024

Expected Output

Leap Year

Question 27

(8 marks)

Problem Name: Product Minus Sum of Digits (variant)

Problem Statement:

Given a positive integer n , compute and return the value (product of digits of n) - (sum of digits of n).

Constraints: $1 \leq n \leq 10^6$

Input / Output format:

First line contains TC. Each of the following TC lines contains a single integer n .
For each test case print the result on a new line.

Example Input 4

123

405

999

560

Example Output

-2

0

524

-11

Question 28 Attempt any one of the following : either option A or option B.

Option A

(12 marks)

Factorial Trailing Zeroes

Given n , print count of trailing zeros in $n!$.

Example

Input: 4

4

10

50

100000

Output:

0

2

12

24999

Option B:

(12 marks)

Number of Common Factors

Given integers a and b, count numbers dividing both.

Example

Input: 4

10 20

16 24

7 49

13 17

Output:

4

4

2

1
